

Line Follower Robot: A Makerspace Lab Project

BY GROUP : E

BATCH : B1

GROUP MEMBERS : HARSHVARDHAN CHOUDHARY (LEADER), GURU PRASATH S L, HILORI JAIN, HIMANSHU, HIMESH JHA, J RAMULU NAIK



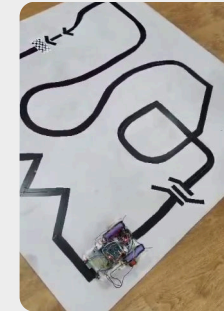
PROBLEM STATEMENT:

Design and develop a line follower robot capable of autonomously navigating a predefined path marked by a contrasting marked line on floor.

The robot should accurately follow the line overcoming challenges such as sharp turns, intersections and orientations along with a special feature .

**Incorporated
special feature :**

**WIRELESS
CHARGING**



The basic principle behind the working involves using combination of infrared sensors and LEDs to detect the contrast between the line and surrounding surface.

Equipment and Components Used :

1) Microcontroller:

Arduino Uno - acts as the brain of the robot

2) Sensors:

Infrared sensors for detecting the line and making decisions, ultra sonic sensor

3) Motor Driver:

L298N - controls the movement of the robot

4) Additional components : Charging module and copper coil

5) Rechargeable Li-ion batteries (3.7V x2) , battery holders and jumper wires

Design and Development Process

BUILDING THE CHASSIS

The first step in creating a line follower robot is building the chassis. This involves constructing the body of the robot, which will hold all of the necessary components. We made this by using acrylic sheets cut by laser.

1

2

DESIGNING WHEELS

Design a 3D model of a wheel using the Autodesk Fusion 360 software keeping in mind the dimensions of motors to fit in. Print it out using 3D printer.

Connecting Motors and Sensors

To bring your line follower robot to life , follow these steps:

1 Motor Connection

Connect the motors to the L298N motor driver onto the Arduino board. Connect the servo motor to motor driver and attach wheels to motor.

2 Sensor Integration

Integrate infrared and ultrasonic sensors with the Arduino Uno for line detection and decision making.

3 Power Management

Ensure the correct connection of rechargeable Li-ion batteries and the use of battery holders for stable power supply.

Programming the microcontroller



The code for line follower include a code covering 4 basics conditions, of 2 sensors, the first one includes both in white so as the code says move forward , if both black then move right and if left one is black and right is white then move left and if right is black and left is white then move right, at the same time we have an initial delay after switching on, so as to refrain sudden start.



Setting up the development environment



Writing and testing the code

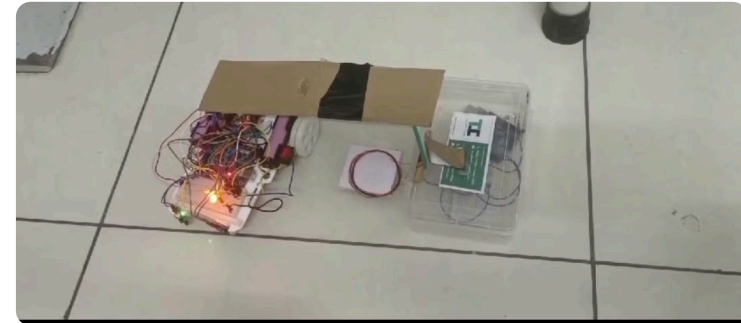


Implementing sensor and motor controls

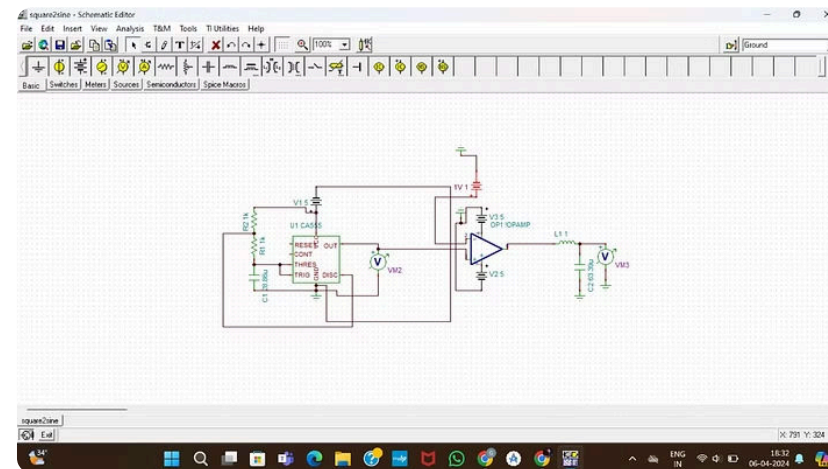
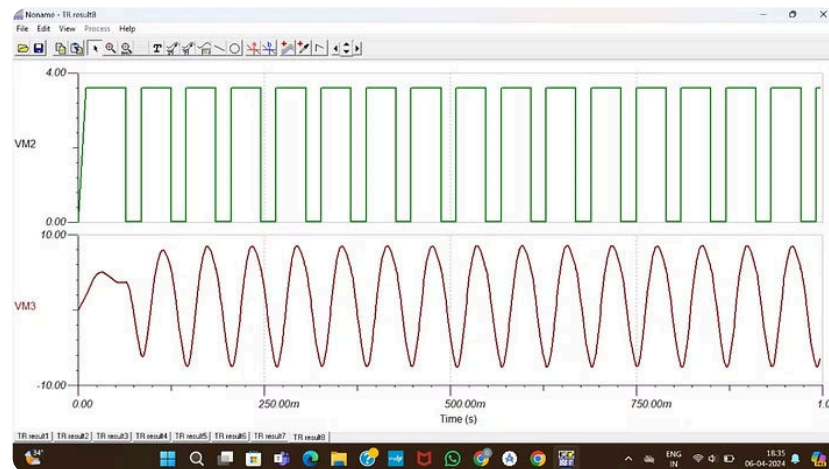
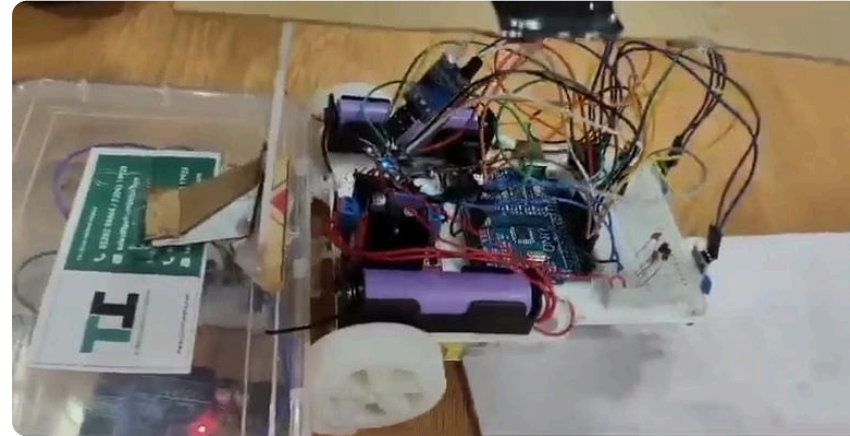
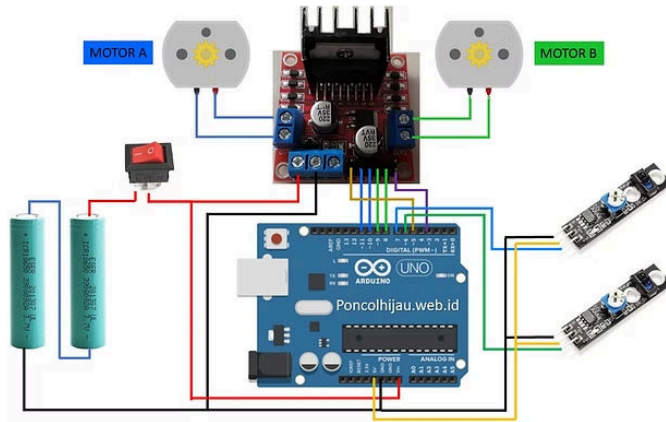
Special Feature: Wireless Charging

NO MESSY CORDS WITHOUT ANY PHYSICAL CHARGING!

The robot's wireless charging system uses advanced electromagnetic induction technology. A futuristic neon glow illuminates the charging area, giving it a modern and innovative feel.



Connection Circuit



Testing and Optimization

Rigorous Testing

The robot undergoes extensive testing on various track layouts and lighting conditions to ensure its reliability.

Continuous Refinement

The team continuously optimizes the robot's performance by fine-tuning the algorithms and hardware integration.

Data-driven Improvements

The team analyzes sensor data and performance metrics to identify and address any issues or areas for improvement.

Conclusion and Future Improvements

1 Successful Project Outcome

The line follower robot project has been a great success, showcasing the students' engineering and problem-solving skills.

3 Continued Learning

The Makerspace Lab course has been an invaluable experience, inspiring the students to continue their pursuit of innovative engineering solutions.

2 Potential Upgrades

The team plans to explore adding more advanced features, such as wireless control and autonomous navigation in complex environments.

